THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** 100 J	Half converted to PE.
3	**(b)** 250 J	mgh.
4	**(b)** 480 W	2400/5.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 3h/4	KE = 1/4 of mgh.
7	**(d)** 4 m/s	sqrt(2 x 10 x 0.8).
8	**(d)** 3 : 20	0.3 : 2.
9	**(a)** Both A and R are true and R is the correct explanation of A.	There is no displacement, so $W = Fs = 0$.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the definition of power does not explain the unit of energy.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	KE change = (1/2)(2)(25 - 9) (1); = 16 J (1).	1+1
12	P = F v (1); F = 10000 N (1).	1+1
13	Work against gravity depends only on the vertical rise (1); PE = mgh (1).	1+1
14	PE at 8 m = 80 J (1); Energy lost = 100 - 80 = 20 J (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	PE lost = 200 J (1); KE gained = 90 J (1); Work against friction = 110 J (1).	1+1+1
16	Energy proportional to x squared (1); Doubling the extension gives 4 times the energy (1); 16 J (1).	1+1+1
17	By force = 90 J (1); By gravity = -60 J (1); KE = 30 J, v = 5.5 m/s (1).	1+1+1
	OR: Total mass 80 kg (1); W = 4000 J (1); P = 400 W (1).	1+1+1
18	Useful P = 3000 W (1); Input = 3000/0.6 (1); = 5000 W (1).	1+1+1
	OR: Initial KE = 600 J (1); Height h = 300/30 = 10 m (1); v = 14.1 m/s (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $PE = mgh = 50 \times 10 \times 10 = 5000$ J (1). (ii) Work $= 5000$ J (1). (iii) $P = W/t = 5000/20 = 250$ W (1). (iv) $P = 5000/40 = 125$ W (1); the work done is unchanged (5000 J), only the rate of doing work changes (1).	1+1+1+2
20	(a) 15 m/s (1); (b) 135000 J (1); Distance 75 m; resistive work 45000 J (1); (c) Engine work 180000 J (d) 18000 W (1); (e) Force = 1200 x 1.5 + 600 = 2400 N; P = 36000 W (1).	1+1+1+1+1
	OR: W by tension = 3000 J (1); PE gain = 2400 J (1); Work against friction = 600 J (1); Efficiency 80 per cent (1); Friction = 50 N (1).	1+1+1+1+1

Mistakes to avoid:

- Saying work is done by holding a load or by pushing a wall; without displacement, work is zero.
- Treating work done by friction or by a force perpendicular to the motion (a coolie walking with a load on his head) as positive.
- Writing $KE = mv^2$ or taking $KE \propto v$; doubling the speed makes the kinetic energy four times.
- Not converting units: minutes to seconds in $P = W/t$, kW to W, or 20 cm of stairs to metres.
- Believing that a machine (lever, ramp) saves work; it only reduces the effort by increasing the distance moved by the effort.
- Using $1\ \text{kWh} = 1000$ J instead of 3.6×10^6 J, or calling kWh a unit of power.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 72 J	F = m(g + a) = 24 N.
2	**(a)** 2520 rupees	360 x 7.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(b)** 2 : 1	v proportional to 1/sqrt(m).
5	**(a)** 2 m/s	P/(mg).
6	**(d)** 400 W	0.8 x 500.
7	**(d)** A is false but R is true.	A is false and R is true.
8	**(d)** -40 J	Negative work.
9	**(c)** A is true but R is false.	$KE = \frac{1}{2}mv^2$ varies as the square of the speed, so R is false.
10	**(d)** A is false but R is true.	The mechanical advantage of a fixed pulley is 1, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Work in = 1000 J (1); Load moves 10 m (1).	1+1
12	Useful power = 500 W (1); t = 5000/500 = 10 s (1).	1+1
13	KE = 2 x 10^6 J (1); d = 40 m (1).	1+1
14	Power is rate; energy = P x t (1); kWh is a convenient large unit (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	v max = sqrt(10) = 3.16 m/s (1); At 0.2 m: PE = 0.4 J (1); KE = 0.6 J (1).	1+1+1
16	PE lost = 35000 J (1); KE gained = 21875 J (1); Energy lost = 13125 J (1).	1+1+1
17	Valley: minimum PE, stable (1); Hilltop: maximum PE, unstable (1); Flat ground: constant PE, neutral (1).	1+1+1
	OR: F x 5 = 100 x 2 (1); F = 40 N, W = 200 J (1); MA = 2.5 (1).	1+1+1
18	PE gained = mgh (1); Machines cannot create energy (1); Extra energy is lost to friction (1).	1+1+1
	OR: W = 600000 J (1); P = 1000 W (1); = 1.34 hp (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $PE = 2 \times 10 \times 20 = 400$ J (1). (ii) $KE = 400$ J, by conservation of energy (1). (iii) $\frac{1}{2} \times 2 \times v^2 = 400$, so $v = 20$ m/s (1). (iv) At 5 m: $PE = 2 \times 10 \times 5 = 100$ J, so $KE = 400 - 100 = 300$ J (1); the total mechanical energy stays 400 J (1).	1+1+1+2
20	Inclined plane: 80 N, MA 5 (1); Lever: effort 100 N, MA 4 (effort arm 1 m) (1); Pulley: 100 N, MA 4 (1); Work output = 400 x height; same (1); Inclined plane has the largest MA here (1).	1+1+1+1+1
	OR: Chemical to heat to steam KE (1); Steam turns turbine, generator makes electricity (1); Overall = 0.8 x 0.4 x 0.95 (1); = 30.4 per cent (1); Rest lost as heat (1).	1+1+1+1+1

Mistakes to avoid:

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- Writing $KE = mv^2$ or taking $KE \propto v$; doubling the speed makes the kinetic energy four times.
- Not converting units: minutes to seconds in $P = W/t$, kW to W, or 20 cm of stairs to metres.
- Believing that a machine (lever, ramp) saves work; it only reduces the effort by increasing the distance moved by the effort.
- Using $1\ \text{kWh} = 1000$ J instead of 3.6×10^6 J, or calling kWh a unit of power.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** the change in KE	Work-energy theorem.
3	**(d)** zero	Perpendicular to displacement.
4	**(a)** 4	MA = load/effort.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(c)** 8 m	0.8 x 10.
7	**(b)** 8 N	Net work 60 = (20 - f) x 5.
8	**(a)** 75 N	300 x 0.4/1.6.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Since the velocity is zero there, all the energy is potential.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the sign of the work does not by itself explain conservation of mechanical energy.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	PE to KE to elastic PE to KE to PE (1); Some energy lost as heat/sound (1).	1+1
12	Work = 60 J (1); v = sqrt(2 x 60/3) = 6.3 m/s (1).	1+1
13	v = 10 m/s (1); p = 50 kg m/s (1).	1+1
14	W = 300 J (1); P = 100 W (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	W = 2000 J (1); P = 1000 W (1); Hold: zero work (no displacement) (1).	1+1+1
16	KE1 = 10000 J; KE2 = 40000 J (1); Work = 30000 J (1); P = 6000 W (1).	1+1+1
17	Hammer's KE is transferred in a very short distance (1); F = W/d is large (1); The force exceeds what hands can give (1).	1+1+1
	OR: KE = 50 J (1); h = 31.25 m (1); At top KE zero (1).	1+1+1
18	Power = 500 x 10 x 20 = 100000 W (1); Electrical = 40000 W (1); 40 kW (1).	1+1+1
	OR: Lift = 225 J (1); Walk: zero work against gravity (1); Displacement perpendicular to weight (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $MA = \frac{\text{effort arm}}{\text{load arm}} = \frac{1.5}{0.5} = 3$ (1). (ii) $E = \frac{L}{MA} = \frac{600}{3} = 200$ N (1). (iii) Load rises $0.6/3 = 0.2$ m (1). (iv) Work by effort $= 200 \times 0.6 = 120$ J; work on the load $= 600 \times 0.2 = 120$ J (1); in a real machine some work is lost against friction, so output work is less than input work (1).	1+1+1+2
20	Total 2250 J (1); KE = PE at 1125 J (1); mgh = 1125, h = 22.5 m (1); v = 21.2 m/s; t = 0.88 s on the way up (1); Total remains constant (1).	1+1+1+1+1
	OR: PE lost = 200 J (1); All converts to heat (1); Friction force x 5 = 200 (1); Force = 40 N (1); Heat produced = 200 J (1).	1+1+1+1+1

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